

Education

University of San Francisco
Bachelor of Science - Physics

Graduated August 2019
Minors in Engineering & Computer Science

Experience

Aqua Finance, Inc. - Remote

February 2026 - Present

Manager of Analytics, Counterparty Risk

- Built counterparty risk frameworks, KPIs, early-warning indicators, predictive models, and dealer scorecards to monitor credit, fraud, operational, and compliance risk across dealers, applications, and sales channels.
- Led fraud detection and dealer monitoring initiatives using structured and unstructured data to identify abnormal activity, emerging risk trends, and escalation priorities.
- Defined monitoring rules, risk thresholds, alerts, and executive dashboards supporting proactive dealer oversight, portfolio health reviews, and risk-committee decision-making.
- Partnered cross-functionally with Credit, Sales, Underwriting, Product, Data, and Engineering teams to translate analytical findings into actionable risk controls and business recommendations.

Forbright Bank - Remote

September 2025 - February 2026

Sr. Analyst - Servicing Analytics

- Delivered analytics, reporting, and strategic insights across Servicing, Collections, Capital Markets, Legal, and Compliance to monitor portfolio health and operations
- Analyzed repayment behavior and servicing trends; designed A/B tests and presented recommendations to leadership, clients, investors, and financial institutions

Solar Mosaic LLC - Remote/Oakland, CA

June 2021 - September 2025

Sr. Risk Analyst

- Developed SQL and Gephi-based network analysis using IP address, geographic, application, and business-registration data to identify relationships among connected entities.
- Built an XGBoost classification model to estimate dealer offboarding risk across multiple prediction horizons, including 3-month, 6-month, and 12-month outcomes.
- Designed analytical datasets and feature pipelines in **Snowflake**, **DBT**, **Apache Airflow** and **Amazon Web Services** to support recurring risk monitoring and model development.
- Built dynamic partner risk scorecards combining model outputs, portfolio performance metrics, and early-warning indicators into structured monitoring frameworks.
- Conducted loan-level performance analysis across delinquency, repayment behavior, credit quality, and portfolio risk indicators to explain deviations from expected outcomes.

L-Egant Solutions, LLC - Irvine, CA

October 2019 - June 2021

Lead Software Developer [Contract]

- Developed Python research software for a second-harmonic dispersion interferometer measuring 2D plasma-jet and gas-jet density profiles.
- Built experimental-control tools for camera configuration, shot-data collection, image capture, calibration workflows, and local data storage.
- Coauthored a peer-reviewed publication in Review of Scientific Instruments

Mechanics Bank Auto Finance - Irvine, CA

August 2019 - June 2021

Data Analyst - Portfolio & Servicing Risk

- Used K-nearest neighbor analysis to compare high- and low-performing collections agents and evaluate patterns in collections behavior and outcomes.
- Built Random Forest models using Equifax, Experian, and TransUnion variables to identify credit-bureau attributes with the greatest lift in predicting delinquent loans.

University of San Francisco - San Francisco, CA *May 2017 - August 2019*
Undergraduate Research Assistant — Plasma Modeling

- Developed a Monte Carlo simulation in Python to model electron transport through argon plasma, tracking particle position, energy, and collision events.
- Delivered a research presentation explaining the simulation methodology and the role of plasma etching in modern semiconductor manufacturing.

Publications

Journal Articles

- [1] C. T. Chavez et al. “Measurement of 2D density profiles using a second-harmonic, dispersion interferometer.” In: *Review of Scientific Instruments* 94.2 (Feb. 2023), p. 023503. ISSN: 0034-6748. URL: <https://doi.org/10.1063/5.0119896>.
- [2] M. Nikolic et al. “Applicability of optical emission spectroscopy techniques for characterization of Ar and Ar/O₂ discharges.” In: *Journal of Physics D: Applied Physics* 54.27 (Apr. 2021), p. 275203. URL: <https://dx.doi.org/10.1088/1361-6463/abf61c>.

Skills

Technical Skills

- Data Engineering & Cloud: DBT, AWS, Snowflake, Docker, Airflow
- Programming & Query Languages: Python, SQL, R, LaTeX, Java
- Scientific Computing & Modeling: Pandas, NumPy, Spark, SciPy, Gephi
- Visualization & Reporting: Tableau, Matplotlib, ggplot2, Mayavi, Streamlit, Plotly, Sigma
- Version Control & Collaboration: Github, Bitbucket